Mini-BEHAVIOR-Gran: Revealing U-Shaped Effects of Instruction Granularity on Language-Guided Embodied Agents

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Abstract

Instruction granularity is an important yet poorly controlled variable in language-guided embodied AI. Existing benchmarks typically pair each task with a single static instruction, making it difficult to study how agent behavior changes when the same task is described at different levels of detail. We introduce MINI-BEHAVIOR-GRAN, a new benchmark for controlled studies of instruction granularity that extends MINI-BEHAVIOR with multiple instruction variants per task, ranging from high-level goal descriptions to step-by-step guidance. Using this benchmark, we compare four candidate metrics for cross-task granularity quantification: token count, entity count, action-verb count, and *planning-width*, and find that width correlates most consistently with agent performance. Using width to organize training and evaluation further reveals a non-monotonic U-shaped relationship between instruction granularity and performance, with peaks at both fine and coarse extremes. Further analysis suggests that the coarse-granularity performance rebound is associated with shallow grounding, where agents learn vision-dominant policies.

1 Introduction

Language-guided embodied agents learn a policy $\pi(a|v, l)$ that maps visual observations v and language instructions l to executable actions a. Recent progress has largely focused on model architectures, most notably the Vision-Language-Action (VLA) paradigm (Kim et al., 2024; Black et al., 2025; Shukor et al., 2025) and its advanced variants that adapt *world models* for future frame prediction ($v_t \mapsto v_{t+1}$) (Wu et al., 2026; Zhao et al., 2025; Bi et al., 2025) or *Chain-of-Thought* (CoT) reasoning for intermediate planning (Ye et al., 2025; Lu et al., 2025; Yin et al., 2025). Yet the impact of *instruction granularity* remains underexplored. Instruction granularity refers to the level of detail provided in l to guide agent decision-making (Arumugam et al., 2017), and it is typically an uncontrolled variable in current datasets and evaluations. It matters because varying granularity fundamentally alters the cross-modal alignment difficulty and the latent planning burden imposed on the agent. Understanding this impact is crucial not only for optimizing instruction design during training, but also for ensuring model robustness across diverse human instruction styles encountered at test time.

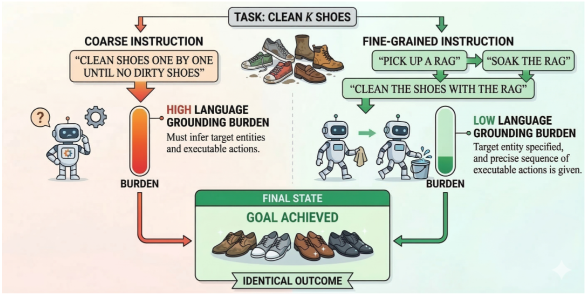

Figure 1: For the same task (cleaning k shoes), a coarse instruction requires the agent to infer target entities and executable actions. In contrast, a fine-grained instruction provides more detailed steps, significantly reducing the language grounding effort to reach the goal.

A systematic investigation of instruction granularity has been hindered by two key bottlenecks. First, existing embodied AI benchmarks lack controlled variation in instruction granularity. Typically, each task instance is paired with a single, static instruction, making it difficult to study how an agent's performance scales as instructions shift from fine-grained (e.g., detailed step-by-step) to coarse (e.g., high-level goal descriptions). For example, popular benchmarks such as LIBERO and BEHAVIOR-1K provide only fixed or high-level instructions. Even recent suites like LIBERO-PLUS (Fei et al., 2025), which aim to increase diversity, primarily vary visual conditions (e.g., lighting, sensor noise) while keeping the language input fixed.

Second, and perhaps more fundamentally, the field lacks a well-accepted and formalized metric for quantifying *instruction granularity*. Since granularity represents the density of guidance provided for decision-making, an ideal metric should directly reflect the complexity of grounding language into executable actions. With the advent of Large Language Models (LLMs), a seemingly natural proxy is to use the hierarchical depth of an instruction, i.e., its specific level within a decomposed plan, as a measure of granularity (Gao et al., 2025). However, such a metric fails to support meaningful cross-task comparisons because it conflates instructional detail with the task's inherent complexity. Different tasks exhibit fundamentally different internal structures: some involve interleaved subgoals that resist clean hierarchical decomposition (e.g., the Sussman anomaly; Sussman, 1975), whereas others can be arbitrarily elongated into superficial substeps. Consequently, hierarchical levels cannot establish a consistent granularity scale across diverse tasks, thereby undermining their utility for investigating the general impacts of granularity on language grounding.

To enable systematic study of how instruction granularity shapes language-guided embodied agents, we introduce MINI-BEHAVIOR-GRAN, an extension of MINI-BEHAVIOR (Jin et al., 2023) that provides multiple instruction variants per task, spanning from high-level goal description to step-by-step instructions. Leveraging Minigrid's engine (Chevalier-Boisvert et al., 2023), we construct 803 unique problem instances across 20 domains, yielding 10,253 trajectory episodes, each paired with dynamic instructions that evolve as the agent progresses through the task. Within this controlled setting, we compare four metrics for quantifying granularity: token count, entity count, action-verb count, and *width*. Unlike *decomposition depth*, *width* measures the max state features jointly tracked per instruction, thereby enabling cross-task comparability of planning complexity (Bonet et al., 2019; Drexler et al., 2024).

Our study shows that *width* correlates more consistently with agent performance across task horizons and VLA variants than other metrics, serving as a better quantitative proxy for instruction granularity. Organizing training by width reveals a U-shaped pattern. Probing this effect via instruction predictability ($P(l|v)$) suggests that the rebound at coarse end is associated with shallow grounding, where agents prioritize single-modality cues and shift toward vision-dominant policies.

Contribution. We make the following contributions: **(1)** Introduce MINI-BEHAVIOR-GRAN, a benchmark for controlled study of instruction granularity. **(2)** Demonstrate that *width* is the most consistent cross-task proxy among the metrics we evaluate. **(3)** Unveil a non-monotonic U-shaped relationship between granularity and agent performance; and **(4)** Provide evidence that the coarse-granularity performance rebound is linked to shallow grounding.

2 MINI-BEHAVIOR-GRAN: A Controlled Benchmark for Instruction Granularity

Following prior work on varying granularities and abstraction in robot instructions (Arumugam et al., 2019), we define *instruction granularity as the level of details and abstraction with which a language instruction specifies a task. In embodied settings, instruction granularity determines the latent planning complexity that an agent must resolve to map instructions into executable actions.*

Reiterating the example in Figure 1, we note that a coarse instruction (e.g., "clean the shoes") leaves the agent with a large language grounding gap, as it must infer the target entities (e.g., which shoes to clean) and the executable actions (e.g., how to clean). In contrast, a fine-grained instruction provides detailed steps that significantly reduce the language grounding effort to fulfill the same task. In this section, we describe how we construct MINI-BEHAVIOR-GRAN to systematically vary instruction granularity for the same task.

2.1 Shared Feature Space and Symbolic Layer

Many embodied AI simulators expose symbolic representations alongside raw pixel observations for task specification, progress monitoring, or success evaluation (Shridhar et al., 2020; Puig et al., 2020; Li et al., 2022a,b). We leverage this symbolic backbone to define a shared feature space Φ common across all tasks, representing each environment state s as a feature vector $\phi(s) = (\phi_1(s), \ldots, \phi_n(s))$. Features in Φ include both Boolean variables (e.g., H_r: holding the rag; S_r: the rag is soaked) and numerical variables (e.g., p_r: distance to the rag; N_f: number of uncleaned shoes). This shared feature space provides the underlying vocabulary for representing instructions, which in turn allows us to quantify instruction granularity across tasks.

2.2 Rule-Based Instruction Representation

Building on the shared feature space Φ, we construct each instruction as a set of *"if-then" rules* that specify desired state progression. This choice is motivated by the fact that sequential decision-making tasks in automated planning are commonly modeled in terms of state variables and transitions with preconditions and effects, following the STRIPS formalism (Haslum et al., 2019). Thus, rule-based instructions explicitly encode the latent planning complexity an instruction imposes, as agents must determine how to transition from states satisfying a rule's precondition to those satisfying its effect. Concretely, an instruction in this study is a set of rules $\mathcal{R} = \{r_1, \ldots, r_m\}$, where each rule r_i is of the form $C_{r_i} \mapsto E_{r_i}$:

- C_{r_i} (the *condition*) is a conjunction of feature constraints. For a Boolean feature p, a constraint is p or $\neg p$; for a numerical feature n, it is $n = 0$ or $n > 0$.
- E_{r_i} (the *effect*) is a conjunction of feature changes. For a Boolean feature p, an effect is p, $\neg p$, or $p?$ (allowing any change); for a numerical feature n, it is $n\downarrow$ (decrease), $n\uparrow$ (increase), or $n?$ (any change).

For each task, we begin by manually constructing a finest-grained rule set $\mathcal{R}^{(0)}$ that decomposes the task into atomic steps, i.e., each rule typically can be fulfilled by a single action. We use the task of *cleaning k shoes* as a running example. Table 1 lists the relevant features from the shared space Φ used in this task.

$\mathcal{R}^{(0)}$ contains over a dozen atomic rules. For brevity, we show representative rules from each phase (full set in § G).

Representative rules from $\mathcal{R}^{(0)}$ (Cleaning Shoes).

$\{\neg H_r,\ p_r > 0\} \mapsto \{p_r\downarrow\}$	(approach rag)
$\{H_r,\ \neg T_s,\ p_s = 0\} \mapsto \{T_s\}$	(turn on tap)
$\{H_r,\ T_s,\ p_s = 0\} \mapsto \{S_r\}$	(soak rag)
$\{S_r,\ N_f > 0,\ p_f > 0\} \mapsto \{p_f\downarrow\}$	(approach shoe)
$\{S_r,\ N_f > 0,\ p_f = 0\} \mapsto \{N_f\downarrow,\ \neg H_r\}$	(clean and drop)
$\{N_f = 0,\ \neg O_r,\ H_r\} \mapsto \{\neg H_r,\ O_r\}$	(put away rag)

A coarse rule is not created by mechanically concatenating the preconditions and effects of its constituent atomic rules. Instead, we treat each merge as an abstraction of a contiguous subplan segment into a macro-rule. The macro-rule exposes only the *net input/output interface* of that segment: its condition specifies when the segment applies, and its effect captures the overall progress achieved. Features that can be consumed entirely within the segment are thus omitted from the expression. As shown in Figure 2, the $\mathcal{R}^{(2)}$ rules for picking up the rag (r_{1a}), soaking it (r_{1b}), and cleaning one shoe (r_{1c}) form a coherent subplan r_1 that abstracts into a single coarse rule in $\mathcal{R}^{(3)}$, which captures the net progress of cleaning one shoe while hiding the intermediate features H_r and S_r.

Feature	Description
H_r	holding the rag (Boolean)
p_r	distance to the nearest rag (numerical)
p_s	distance to the sink (numerical)
p_f	distance to the nearest uncleaned shoe (numerical)
N_f	number of uncleaned shoes (numerical)
T_s	tap (sink) is turned on (Boolean)
S_r	rag is soaked (Boolean)
C_i	shoe i is cleaned (Boolean)
O_r	rag is on the floor (Boolean)

Table 1: Features used in the Cleaning Shoes task.

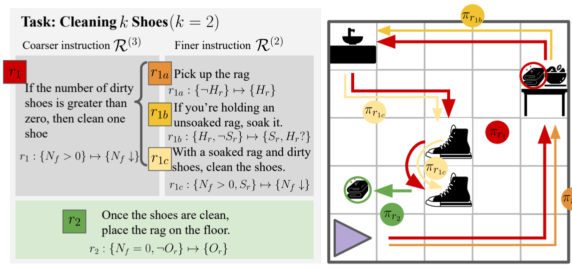

Figure 2: A running example of constructing coarse-grained instructions by merging atomic rules into macro-rules. Features H_r and S_r are omitted as they can be consumed entirely within the subplan chunk.

After each merge, we verify the semantic validity of the resulting coarse rules through expert review and use a planning solver to ensure every $\mathcal{R}^{(k)}$ is well-formed: sequentially reachable and goal-terminating (Bonet and Geffner, 2021).

2.3 Dynamic Instruction Generation

To communicate these rules to the agents, we translate each rule into a natural-language sentence via a template-based generator. The templates verbalize the feature conditions and desired effects in each rule. For example, $\{H_r, \neg S_r\} \mapsto \{S_r\}$ is realized as "If you are holding the rag but the rag is not yet soaked, soak the rag."

The benchmark supports *dynamic* instruction generation that updates with task progress. As illustrated in Figure 3, during both training and evaluation, an instructor module monitors the simulator state at each timestep, computes the set of applicable rules $\mathcal{R}_a$, and determines which rule r_t

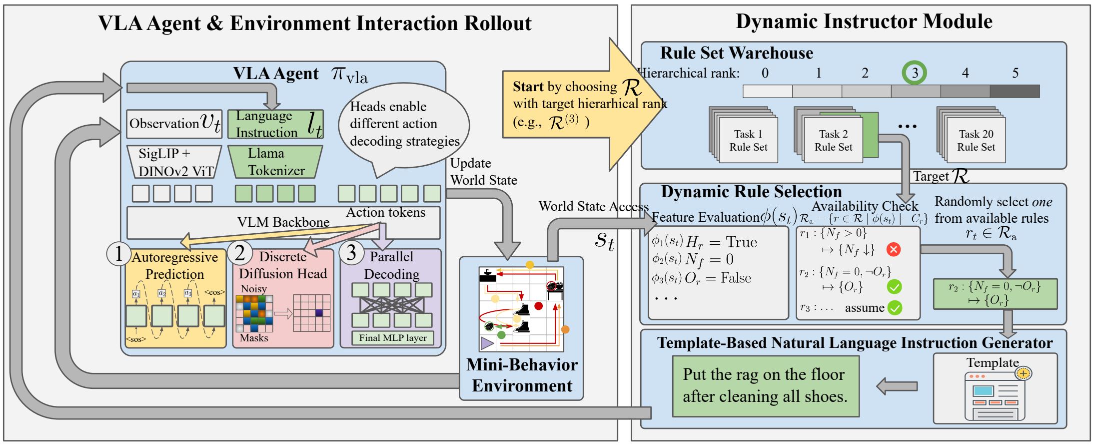

Figure 3: MINI-BEHAVIOR-GRAN Experimental Pipeline. We evaluate the impact of instruction granularity across three VLA action-decoding strategies. During inference time, a dynamic instructor evaluates the current world state to decide available rule set $\mathcal{R}_a$ and active rule r_t. A rule-based natural language instruction l_t is generated accordingly. The process iterates until completion or max steps.

Rank	N	$\overline{\text{Len}}$	U_{tok}	$\overline{\text{Plan}}$
0	51,474	40.58 ± 10.59	145	83.43 ± 53.66
1	27,837	38.19 ± 11.53	139	85.81 ± 56.91
2	14,997	33.51 ± 11.17	253	93.68 ± 57.24
3	6,816	34.09 ± 13.99	171	120.26 ± 51.39
4	1,691	41.78 ± 22.67	112	169.32 ± 37.71
5	923	33.31 ± 7.68	85	146.70 ± 23.56

Table 2: Distribution of instruction variants across granularity ranks. **N**: number of instruction instances; $\overline{\text{Len}}$: mean instruction length in words; $\mathbf{U}_{\text{tok}}$: number of unique tokens; $\overline{\text{Plan}}$: mean plan length in steps.

to activate. This setup ensures that the agent always receives state-relevant guidance, in contrast to existing benchmarks (Li et al., 2022b; Fei et al., 2025) that provide only a single static instruction.

2.4 Benchmark Analysis

Table 2 summarizes the scale and coverage of MINI-BEHAVIOR-GRAN. Across 20 domains, we define a shared feature space Φ with 166 grounded features. Task-specific feature usage ranges from only 2 features in simple tasks such as *Opening Packages* to 29 in long-horizon tasks such as *Cleaning Up the Kitchen*. Despite their structured origin, the resulting instructions are of moderate linguistic complexity, averaging 40.2 words in length (see § F.3 for details).

3 Quantifying Instruction Granularity

While hierarchical rank remains a useful within-task measure, it cannot be used as a consistent cross-task metric to study how instruction granularity impacts agent performance across a diverse task set. The reason is that the number of attainable levels depends on the internal structure of each task: some tasks admit long chains of cleanly mergeable subgoals, whereas others involve interleaved dependencies that prevent further merging. Consequently, the same hierarchical rank across two different tasks may entail vastly different language grounding complexities. We therefore seek a separate cross-task metric for quantifying granularity.

3.1 Candidate Metrics

To this end, we evaluate four candidate metrics, each corresponding to a different hypothesis about which properties of an instruction plausibly reflect its granularity and associated grounding burden. *Token Count* measures *verbal elaboration*: finer-grained instructions often require more words to spell out procedural details. *Entity Count* measures *referential grounding load*, since mentioning more objects or locations requires the agent to ground a larger set of task-relevant referents. *Action-Verb Count* measures *procedural complexity*, as finer-grained instructions tend to specify more individual actions. While these three metrics capture surface-level properties, our fourth metric, *Width*, captures a more structural notion of planning burden.

3.2 Instantiating Width as a Metric

Among the candidate metrics we consider, *width* is the least surface-oriented. Originating from width-based planning (Lipovetzky and Geffner, 2012, 2017), for a rule r_i, its width $w(r_i)$ is the maximum number of features that must be simultaneously tracked along an optimal transition from a state sat-

State Progression	Concurrent Features Tracked	Width
Initial State	$[\langle\rangle]$	0
Navigate to rag	$[\langle p_r = X\rangle, \ldots, \langle p_r = 0\rangle]$	1
Pick up rag	$[\langle p_r = 0\rangle, \langle H_r\rangle]$	1
Navigate to sink	$[\langle H_r, p_s = X\rangle, \ldots, \langle H_r, p_s = 0\rangle]$	2
Turn on tap (sink)	$[\langle H_r, p_s = 0\rangle, \langle T_s\rangle]$	2
Soak rag	$[\langle H_r, T_s, p_s = 0\rangle, \langle S_r\rangle]$	**3**
Navigate to Shoe 1	$[\langle H_r, S_r, p_{f1} \downarrow\rangle, \ldots,]$	**3**
Clean Shoe 1	$[\langle H_r, S_r, p_{f1} = 0\rangle, \langle C_1\rangle]$	3
Navigate to Shoe 2	$[\langle H_r, S_r, C_1, p_{f2} \downarrow\rangle, \ldots]$	**4**
Clean Shoe 2	$[\langle H_r, S_r, C_1, p_{f2} = 0\rangle, \langle C_1, C_2\rangle]$	4
Put rag on floor	$[\langle C_1, C_2, H_r\rangle, \ldots, \langle C_1, C_2, O_r\rangle]$	3

Table 3: State progression and width for the "Cleaning k Shoes" ($k = 2$). Width scales as $w = k + 2$.

isfying C_{r_i} to one satisfying E_{r_i}. This differs from simply counting the number of features mentioned in the rule's condition or effect, since these features need not all be coordinated concurrently. We instantiate instruction-level width by aggregating over its constituent rules $w(\mathcal{R}) = \max_{r_i \in \mathcal{R}} w(r_i)$. A special case of $w = 0$ occurs for situations where the desired state progression can be achieved in zero or one step (Bonet and Geffner, 2024); this will appear in our experimental analysis.

Table 3 illustrates width computation for a two-shoe cleaning task. Approaching the rag ($p_r \downarrow\to p_r = 0$) requires tracking only p_r ($w = 1$). Obtaining the rag at $p_r = 0$ ($\to H_r$) also needs only p_r ($w = 1$). Then, maintaining H_r while progressing raises width to 2. Soaking the rag requires simultaneously maintaining three features: at sink ($p_s = 0$), holding rag (H_r), and tap on (T_s), yielding $w = 3$. Cleaning the first shoe (C_1) requires tracking H_r, S_r, and p_f ($w = 3$). Cleaning the second shoe adds maintaining C_1, increasing width to 4. We acknowledge that w is not a direct measure of internal grounding difficulty, since learning-based embodied agents operate as black-box decision-makers rather than explicit width-based search. However, w provides a cross-task proxy for the latent planning demands. We thus view it as a planning-inspired measure of granularity.

4 Experiments

4.1 Experimental Setup

Data. We use MINI-BEHAVIOR-GRAN, which contains 20 long-horizon household tasks defined over a shared feature space Φ. For each task, the benchmark provides 50 training and 10 evaluation instances, each paired with reference trajectories generated by the BFWS planning solver (Lipovetzky and Geffner, 2017).

Action-Decoding Archetypes. The primary architectural difference among recent VLA models lies in their action-decoding strategy (Zhong et al., 2025). We therefore adopt a unified VLM backbone and instantiate three representative action-decoding paradigms (see the left panel of Figure 3): (1) **Autoregressive** (AR), exemplified by RT-2 (Zitkovich et al., 2023), OpenVLA (Kim et al., 2024), and related models (Reed et al., 2022; Huang et al., 2024; O'Neill et al., 2024), which predict actions sequentially via causal attention; (2) **Discrete Diffusion**, which predicts action chunks through a diffusion process (Black et al., 2025; Bjorck et al., 2025; Shukor et al., 2025; Liang et al., 2025); and (3) **Parallel Decoding**, which uses bidirectional attention for single-pass action prediction (Kim et al., 2025b; Song et al., 2025; Yin et al., 2025; Wang et al., 2025; Xiao et al., 2025). Implementation details are given in § H.

Two-Stage Experimental Protocol. We conduct experiments in two stages. In **Stage I** (§ 4.2), we keep the training distribution fixed and train all models under a mixed-granularity regime that samples from all available instruction variants. This stage is used only to compare candidate granularity metrics by how consistently they correlates with agent Success Rate (SR) across tasks, task horizons, and action-decoding architectures. In **Stage II** (§ 4.3–§ 4.4), after identifying which is the most consistent metric, we use it to partition instruction variants into *Fine*, *Medium*, and *Coarse* groups and study how different granularity mixtures affect learning and generalization. Specifically, we consider seven training settings: **Pure F/M/C** (1 : 0 : 0), (2) **Centroid** (1 : 1 : 1): uniformly sample from all three classes; (3) **Axial F/M/C** (3 : 1 : 1): emphasizes one class with some exposure to the others.

4.2 RQ1: Which metric best captures instruction granularity?

We first ask which candidate metric most consistently captures instruction granularity in a way that it reflects embodied language grounding difficulty across tasks. We train all models under a balanced granularity regime and evaluate the Pearson correlation between each candidate metric and agent SR. A useful cross-task metric should exhibit a stable relationship with performance across tasks of different sequential burdens.

Why partition by instruction horizon? A critical confound in evaluating instruction granularity

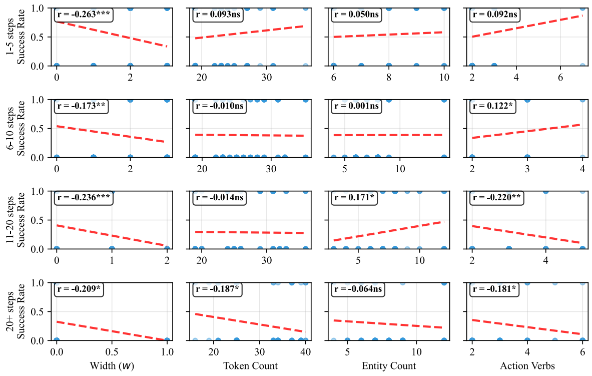

Figure 4: Metric-success correlation by task-horizon, with *width* showing consistent negative correlations.

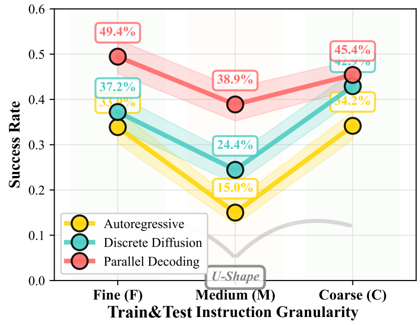

Figure 5: In-Distribution (matched train/test) SR across granularities (i.e., train Fine → test Fine, etc). A non-monotonic U-shaped trend appears.

is sequential burden. A natural choice would be to stratify examples by the total number of primitive actions in the expert trajectory. However, this is not the most relevant notion of difficulty in our setting. In MINI-BEHAVIOR-GRAN, agents do not ground a single static instruction over the entire trajectory; instead, the instructor issues a sequence of rule-conditioned instruction updates as execution unfolds. The more relevant notion of sequential burden is therefore the *instruction horizon*, defined as the number of rule calls issued during execution. We thus partition metric–SR associations evaluation instances by instruction horizon.

Width vs. surface-level metrics. We compare between token count, entity count, action-verb count, and width. Figure 4 show that width is the only metric that exhibits a statistically significant and consistently negative correlation with SR across all instruction-horizon bins (e.g, -0.263 in the 1-5 steps bin). In other words, as width increases, agent success decreases in a stable manner regardless of sequential burden. By contrast, the other three metrics are notably less consistent. Token count is insignificant in the first three bins and becomes significantly negative only in the longest-horizon bin. Entity count is mostly insignificant and even becomes weakly positive in the 11–20 bin. Action-verb count reaches significance in several bins, but its direction flips across horizons, indicating that it does not provide a stable notion of granularity across tasks. We thus use width to organize training in the remainder of the paper.

4.3 RQ2: How does agent performance vary across instruction granularity?

Since RQ1 identifies width as the most consistent cross-task proxy for instruction granularity, we use it to organize the remaining experiments. To enable tractable analysis while maintaining statistical power, we partition instruction variants into three granularity groups based on width: *Fine* ($w = 0$), *Medium* ($w = 1$), and *Coarse* ($w \geq 2$). This grouping is necessary because higher widths ($w \geq 3$) have fewer instances and appear in fewer domains, making per-width analysis underpowered for cross-training comparisons. However, to ensure this grouping does not obscure finer-grained trends, we present a per-width breakdown of performance in Table 5. Figure 5 reveals a consistent non-monotonic U-shaped relationship between instruction granularity and agent performance across all three action-decoding archetypes. At the fine extreme ($w = 0$), where instructions require minimal latent reasoning, models achieve high success rates. Performance then drops at medium granularity ($w = 1$), before rebounding under coarse instructions ($w \geq 2$). This is notable because width tracks latent coordination burden: if granularity affected performance only through increasing planning difficulty, one would expect a monotonic decline rather than a recovery at the coarse end.

While our main analysis groups $w \geq 2$ as *Coarse*, examining per-width performance reveals a graded structure in this regime. Table 5 shows that SR at $w = 2$ and $w = 3$ declines progressively, collapsing entirely at $w \geq 4$. Nevertheless, the overall *Coarse* group still outperforms *Medium* ($w = 1$) when aggregated across tasks. This aggregation is dominated by $w = 2$ (which appears in 19 tasks), while the lower-performing $w = 3$ and $w \geq 4$ appear in only 10 and 3 tasks respectively.

4.4 RQ3: What drives the U-shaped effect?

The coarse-granularity rebound is initially counter-intuitive: if width reflects latent coordination burden, why does performance improve again when instructions become coarser? We hypothesize that this rebound stems from increasing instruction predictability $P(l|v)$, which makes coarse instructions easier to infer from visual context alone. This high predictability incentivizes the policy to exploit single-modality cues, leading to a dominant reliance on visual features rather than complex cross-modal coordination.

This increased predictability has a formal basis: coarse instructions subsume multiple finer-grained variants. Formally, consider a surjective mapping f from fine-grained labels l_i to coarser ones c_j. The conditional probability of a coarse label aggregates its fine-grained constituents: $P(c_j|v) = \sum_{l_i \in f^{-1}(c_j)} P(l_i|v)$. This aggregation makes coarse instructions statistically easier to predict from vision alone.

We confirm this empirically by training a VLM-based instructor (Qwen3-VL-4B with LoRA) to predict instructions from visual observations. As Figure 6 shows, cross-entropy loss decreases monotonically with coarseness (Fine: 0.078, Medium: 0.063, Coarse: 0.047), a relative drop of 40% from fine to coarse, confirming that $P(l|v)$ indeed increases with coarseness.

We next ask whether this increased predictability changes how much models actually rely on language. Following prior work (Fei et al., 2025), we use ΔSR when language is weakened or removed as a probe of grounding reliance. When full instructions are removed entirely, fine-trained policies suffer a severe 36.3% drop, whereas coarse-trained policies show a much smaller 20.0% gap (Figure 7). Notably, under reduced informativeness (*w/ goal lang*), the Pure-C policy exhibits only a negligible 2.8% advantage over having no language at all (15.6% vs. 12.8%). This indicates that policies trained on coarse instructions develop shallow grounding: language is not deeply integrated into decision-making.

Table 4 provides a *supplementary diagnostic*, for the shallow-grounding account. Using a layer-wise Binomial test ($H_0 = 0.5$), we find that Discrete Diffusion models exhibit a shift consistent with greater visual reliance at the coarse end (T2, $p = 0.01$; T3, $p < 0.01$). AR models show a similar but weaker pattern: a stronger language grounding phase from F to M granularity (T1: 25/32, $p \approx 0.001$), followed by a directional shift toward visual reliance at the coarse end (T3: 21/32). We therefore interpret these results as converging evidence rather than definitive mechanistic explanation. Note that T2 and T3 are not redundant: action tokens can also attend to previous action tokens, so a decrease in language attention does not imply a one-to-one increase in visual attention.

Together, these results reveal a three-regime account of the U-shaped performance pattern. At **fine granularity** ($w = 0$), Low $P(l|v)$ (high CE loss) means language provides unique information. Combined with low-width grounding burden, this drives genuine multimodal alignment for generating actions, yielding high SR. **Medium granularity** ($w = 1$) introduces a moderate increase in grounding burden. The model attempts to ground language but struggles with the increased complexity, leading to the lowest performance. At **coarse granularity** ($w \geq 2$), instructions are too complex to ground effectively, and high $P(l|v)$ incentivizes agents to exploit visual cues. This visual-dominant strategy recovers SR but sacrifices language grounding.

Grounding via Mixed-Granularity Training. We hypothesize that increasing the diversity of the training language space should artificially lower average $P(l|v)$, forcing agents to re-engage with text. Our mixed-granularity training confirms this: all mixed-granularity models exhibit substantially larger ΔSR gaps than their Pure-X counterparts. Among these, the fine-weighted mixture *Axial-F* emerges as the optimal compromise, achieving strong overall performance while maintaining larger ΔSR gaps indicative of language grounding.

Asymmetric Generalization. We observe an asymmetric transfer pattern (Table 5): coarse-trained models successfully zero-shot transfer to fine-grained instructions, but fine-trained models fail on coarse-grained ones. This asymmetry aligns with our ΔSR analysis: coarse training induces a visual-dominant policy, hence refining the instruction yields minimal behavioral change. Fine training induces tight coupling with language, making the policy brittle when detailed guidance drops. Also see § I for failure mode analysis.

5 Related Work

Instruction Abstraction and Granularity. Here, *instruction granularity* refers to the level of detail that affects the latent planning burden of an instruction. This differs from *linguistic abstraction*, defined as representational compression (Wing, 2010;

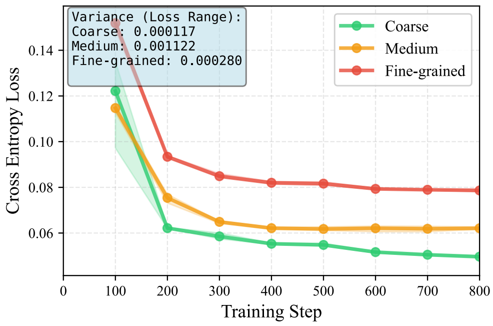

Figure 6: VLM instruction-prediction Cross Entropy (CE) loss across granularities and 3 seeds. Larger CE loss indicates lower $P(l \mid v)$.

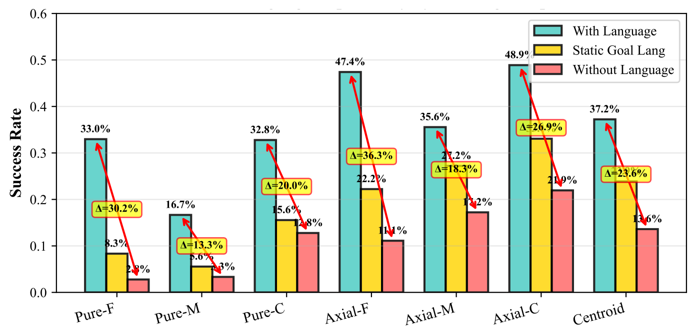

Figure 7: Smaller SR gaps indicate a shift towards shallow grounding. Under semantic perturbation, the Pure-C setting exhibits tiny variance (2.8%) between *w/ goal lang* and *w/o lang*.

Model	T1	T2	T3
AR	78.1% (25/32)**	12.5% (4/32)	65.6% (21/32)
Parallel	15.6% (5/32)	28.1% (9/32)	15.6% (5/32)
Diffusion	53.1% (17/32)	71.9% (23/32)*	75.0% (24/32)**

Table 4: Layer-wise ($N = 32$) attention trends from action tokens. **T1**: lang $\uparrow$ ($F \to M$); **T2**: lang $\downarrow$ ($M \to C$); **T3**: vis $\uparrow$ ($M \to C$). $^*p < 0.05$, $^{**}p < 0.01$ (Binomial test). Parallel models lack consistent trends, likely due to modality entanglement inherent in bidirectional attention (Kim et al., 2021). T2/T3 are not zero-sum as the it includes self-attention. Full table and stats in § I.2.

Training Setup	Width 0 (20 tasks)	Width 1 (20 tasks)	Width 2 (19 tasks)	Width 3 (10 tasks)	Width ≥ 4 (3 tasks)
Pure-F	34.9%	16.9%	10.8%	1.0%	0.0%
Pure-M	12.2%	22.0%	11.7%	0.0%	0.0%
Pure-C	26.3%	33.3%	32.9%	22.9%	0.0%
Axial-F	50.2%	53.3%	48.8%	34.3%	0.0%
Axial-M	44.7%	43.5%	45.8%	29.5%	0.0%
Axial-C	55.7%	52.2%	50.8%	37.1%	0.0%
Centroid (Mix)	42.7%	38.4%	40.8%	28.6%	0.0%

Table 5: Success rate (%) across training setups and test granularities. Numbers in parentheses indicate task coverage at each width. For Pure models, cells where test width was unseen during training measure zero-shot transfer. Transfer is asymmetric: coarse $\to$ fine succeeds.

Lachmy et al., 2022; Zheng et al., 2024). Granularity varies independently: "tidy the room" is coarse yet non-abstract (underspecified actions, no compression); "repeat wiping until clean" is specific yet abstract (prescribes action via looped condition).

While not framed as the term "granularity", we acknowledge that a rich body of work has focused on augmenting language specificity and its impact, for example through coarse-to-fine semantic parsing (Dong and Lapata, 2018; Li et al., 2020). Other work generates compositional instructions via constraint dropout and sub-task recombination (Huang et al., 2025b), or construct counterfactual instructions to augment diversity (Glossop et al., 2025).

Varying instruction detail is often handled via a simple two-part decomposition: high-level plans are converted into sequences of low-level steps before being grounded into actions (Arumugam et al., 2017; Yang et al., 2024; Shi et al., 2025). Most recent Vision Language Action (VLA) works, however, operate with static language instructions (Ma et al., 2025; Li et al., 2025). Some models like InstructVLA and ThinkAct explore having chain-of-thought reasoning or planning steps before action decoding (Huang et al., 2022; Yang et al., 2025; Huang et al., 2025a). These hierarchical definitions remain task-specific and qualitative, lacking a unified, cross-task metric to establish shared granularity levels across tasks. Consequently, quantitatively controlled studies on the formal dynamics of instruction granularity remain scarce.

Language Grounding vs. Shortcuts. Prior work highlights a dichotomy between fragile language sensitivity (Akula et al., 2022) and visuo-motor shortcuts (Xing et al., 2025; Zhang et al., 2026). We reconcile this via width (w), providing complementary insights into the trade-offs between language grounding and visual reliance.

6 Conclusion

We introduce MINI-BEHAVIOR-GRAN, the first testbed for studying the impact of instruction granularity. Comparing four metrics, we find *width* most consistently correlates with grounding difficulty. Using width to organize training reveals a U-shaped granularity-performance pattern, driven by a trade-off between instruction predictability ($P(l|v)$) and planning complexity. We demonstrate that mixed-granularity training balances language sensitivity without sacrificing the visual reasoning skills learned from coarse instructions.

7 Limitations

Discrete Action Spaces. We evaluate our framework in environments with discrete action spaces, a common abstraction in language-conditioned embodied AI (e.g., ALFWorld-based agent (Xiong et al., 2025; Kim et al., 2025a)) and LLM-based agent systems (Qin et al., 2024; Patil et al., 2024), where high-level reasoning is decoupled from low-level execution via skill primitives or API calls. This choice is intentional: it isolates symbolic planning from continuous control noise, allowing us to cleanly attribute observed effects to instruction granularity. However, extending this granularity-aware framework to hybrid discrete-continuous or continuous action spaces remains essential for real-world robotics applications, where low-level motor commands interact with high-level linguistic abstractions.

Scope of Granularity Definition. We view subgoal decomposition as the primary axis of instruction granularity, as it directly determines the latent planning burden—the central construct we aim to measure. This focus aligns with the classical planning view that hierarchical task decomposition is the fundamental mechanism for reducing search complexity. Other forms of granularity (e.g., vagueness, abstraction via loops or branching) are important but introduce additional complexities that would confound our initial investigation. By first isolating and understanding the effect of subgoal decomposition in a controlled setting, we establish a foundation for future work to incorporate these other dimensions. Our benchmark and findings provide a baseline against which such extensions can be compared.

Practicality of width computation. Computing width currently requires access to symbolic state features and optimal plans, which limits its direct applicability to natural language instructions in unstructured environments. We emphasize, however, that width is designed as an analytic metric for controlled studies, not as a deployable measure for real-world settings. Within our benchmark, width enables precise quantification of latent planning burden, allowing us to isolate the effect of instruction granularity and uncover fundamental phenomena such as the U-shaped performance curve and asymmetric transfer. Extending width estimation to realistic settings remains future work and does not affect the validity of our findings.

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A Use of LLM Statement

An LLM was used only for polishing language; the scientific content is entirely original.

B Technical Appendix

Appendix Table of Contents